

Sampling Stars from the Sky – an example of random, systematic, and stratified sampling techniques

***From Petocz (1990), Sampling Space: Practical Experiments for Teaching Sampling (<http://www.stat.auckland.ac.nz/~iase/publications/18/BOOK1/A4-9.pdf>)*

“Star light, star bright, first star I see tonight,” ...but the first star of how many? How many stars are there in the sky? The figure on the next page is a star map and is assumed to represent a grid of stars observed in the sky at a given time. While you could just count the stars, let’s employ statistics to help us out with the task. Use a random sample to determine how many stars there are in the sky. Please answer the following to accomplish this task:

- Explain the method you would use (using a table of random digits or a computer program) to select a random sample of five squares from the 100 listed in the grid (10 rows x 10 columns). Note that $N=100$, $n=5$.
- Using your method, select a random sample of 5 squares. Which five squares did you select?
- How many stars does each grid have? What is the total number of stars in your sample?
- Multiply this number by $100/5$ (or 20) to estimate the total number of stars in the sky. We have sampled $1/20$ ($5/100$) of the total. To find the total we multiply our result by the reciprocal of the portion we have sampled.

Other sampling methods can be used with this example as well:

- **Systematic sampling.** Every k^{th} observation is chosen, where k is approximately N/n . In our example above, $k=100/5 = 20$. Systematic sampling is easier to conduct than simple random sampling, but can be more likely to produce biased estimates. Sampling stars example – Pick a random number from 1 to 20, then choose every 20th square (ex, if your random number is 17, choose 17th, 37th, 57th, 77th, and 97th squares).
- **Stratified sampling.** Group squares by whether they are part of a nebula or void. For example, if there are 40 squares which are part of a nebula and 60 squares part of a void, then randomly choose 2 nebula squares and 3 void squares to estimate the total number of squares.

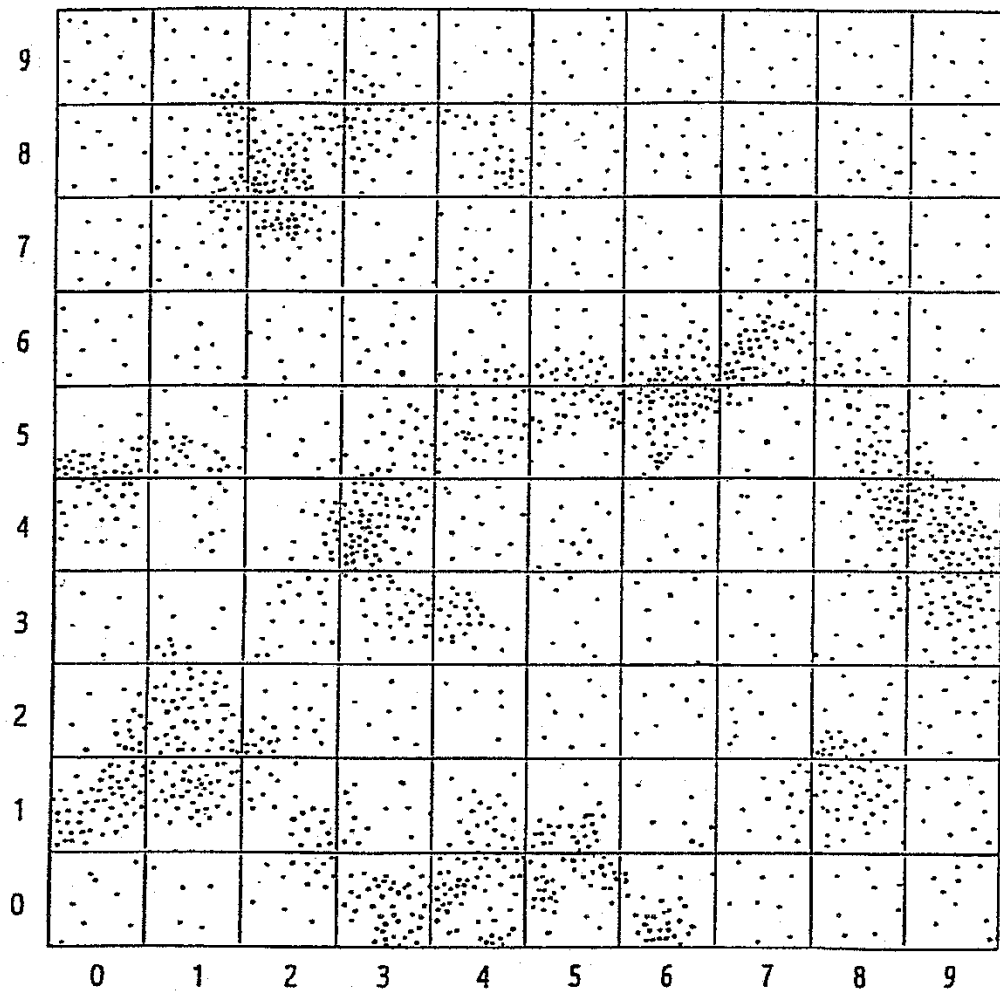


FIGURE 1
Sample space star map